

Length Measurement — Inches and Centimeters

25 practice questions · Little Newtons Class 9 · Grades 1–2 · Print-and-solve packet

Read each question with your child. Encourage them to estimate first, then measure or answer. Full explanations and answers are on the SOMATH blog post:

schoolofmath.us/posts/length-measurement-inches-centimeters-little-newtons.

Question 1 — reading a ruler in inches

A crayon starts at the 0 on an inch ruler and its right end lines up with the number 4. How long is the crayon?

Question 2 — reading a ruler in centimeters

A block starts at the 0 on a centimeter ruler and its right end lines up with the number 6. How long is the block?

Question 3 — the 0-mark trap

Emma measures a pencil. She lines up the left end of the pencil with the LEFT EDGE of the ruler (not with the 0). The right end lines up with the 7. What is wrong with Emma's measurement?

Question 4 — between two numbers, grade 1 rounding

A worm on an inch ruler starts at 0 and ends between 2 and 3, closer to 3. Grade 1 rounding: how long is the worm?

Question 5 — grade 2 half-inch reading

A leaf on an inch ruler starts at 0 and ends exactly halfway between 5 and 6. Grade 2: how long is the leaf?

Question 6 — picking the unit

A teacher is measuring the length of a soccer field. Should she use inches or a different unit?

Question 7 — matching object to unit

Match each object to the best unit for measuring it:

(a) a paperclip (b) a bedroom (c) a pencil (d) a school hallway

Units: inches / feet

Question 8 — estimating with a reference

A crayon is about 3 inches. A pencil looks about twice as long as the crayon. About how long is the pencil?

Question 9 — estimating in centimeters

A pinky fingernail is about 1 cm wide. A key looks about 5 pinky fingernails long. About how long is the key?

Question 10 — inch vs. cm on the same object

Sam measures the same book in both units. He gets 8 in and 20 cm. Which number is bigger — and why?

Question 11 — comparing two lengths

Maya's marker is 5 in long. Ben's marker is 8 in long. Whose marker is longer, and by how much?

Question 12 — comparing in centimeters

A red ribbon is 12 cm. A blue ribbon is 7 cm. How much longer is the red ribbon than the blue ribbon?

Question 13 — adding lengths

A pencil is 6 in and its cap is 2 in. If you snap them together end-to-end, how long is the pencil-plus-cap?

Question 14 — three lengths added

A train of blocks is made of a red block (4 cm), a green block (3 cm), and a blue block (5 cm). How long is the whole train?

Question 15 — missing length

A pencil-plus-cap together is 9 in long. The cap is 2 in. How long is the pencil by itself?

Question 16 — measuring in different units on the same object

The same eraser is measured as 2 in and about 5 cm. Both answers are correct. Why?

Question 17 — unit label matters

Aiden wrote "My shoe is 9." Why is this answer incomplete?

Question 18 — word problem with addition

Sara has a rope that is 7 ft long. She ties on another rope that is 4 ft long. How long is her rope now?

Question 19 — word problem with subtraction

A yarn is 15 cm long. Nina cuts off 6 cm. How much yarn is left?

Question 20 — comparing to the child

Lucas is 48 in tall. His little sister is 39 in tall. How much taller is Lucas than his sister?

Question 21 — estimation catches a mistake

Nora estimates her pencil is about 7 in. She measures with a ruler and writes down 25 in. What probably happened?

Question 22 — count the spaces, not the tick marks

A ruler shows tick marks at 0, 1, 2, 3, 4, 5, 6. How many INCHES is the ruler? How many tick marks are there?

Question 23 — which side of the ruler?

Grace's ruler has inches on top and centimeters on the bottom. She needs to measure a bookmark in centimeters. Which side does she use?

Question 24 — challenge: two-step problem

A rope is 20 in long. It is cut into two pieces. The first piece is 8 in. How long is the second piece?

Question 25 — challenge: estimation + measurement together

Your desk is about as long as 3 pencils lined up end-to-end. If one pencil is 7 in, about how long is the desk?

Want the answers and click-to-reveal explanations? Every question has a hidden step-by-step solution on the blog post. Read the full lesson at schoolofmath.us.

Free 30-minute evaluation. Not sure whether your child is on track for Little Newtons, Kid Einsteins, or something else? Book a free in-person evaluation at 226 W 79th Street and we'll send a written diagnostic within 48 hours — yours to keep whether you enroll or not. schoolofmath.us/evaluation or call (646) 668-6151.